GG–C–133
APRIL 27, 1956

SUPERSEDING
Int. Fed. Spec. GG–C–00133(VA)
Revision—1
March 24, 1955

FEDERAL SPECIFICATION

CASSETTE CHANGER

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope.—This specification is intended to cover a device to permit stereoscopic radiography in the vertical plane.

1.2 Classification.

1.2.1 *Types and classes.*—Cassette changers covered by this specification shall be of the following types and classes:

Type I.—Non-photo-timed
- Class 1.—Without bucky
- Class 2.—With bucky
- Class 3.—With grid

Type II.—Photo-timed
- Class 1.—Without bucky
- Class 2.—With bucky
- Class 3.—With grid

1.2.2 *Styles.*—Cassette changers of type I or type II, shall be of the following styles:
- Style I.—Floor mounting
- Style II.—Wall mounting

2. APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 There are no other specifications and standards applicable to this specification, except those referenced in section 7 by individual departments.

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

3. REQUIREMENTS

3.1 General.—It shall be possible to convert all type I, class 1, cassette changers supplied under this specification to type II, class 2 or class 3 at a later date if so desired. Cassette changers supplied under this specification, shall be designed to permit such modification.

3.2 Components.—The unit shall consist of the following component parts:
- (a) Supporting frame
- (b) Shifting mechanism
- (c) Immobilizing device
- (d) Bucky diaphragm (for type I, class 2 and type II, class 2).
- (e) Photo-timer receiver (for type II).
- (f) Grid (for type I, class 3, and type II, class 3).

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3.2.1 *Supporting frame.* — The supporting frame shall be of such design and construction to rigidly support the shifting mechanism with any attachments referred to in the bid invitation and covered by this specification.

3.2.1.1 *Mounting.*—Supporting frames shall be such design as to permit the easy positioning of a wheel chair with a patient on it, for chest radiography. The supporting frame shall also be of such design as to permit chest radiography of a patient seated in a wheel chair and be provided with sufficient knee clearance for the patient.

3.2.1.2 Upon the supporting frame shall ride the cassette shifting mechanism. Provision shall provide for the vertical movement of this mechanism from a lowest position when the chin rest of the exposure compartment would be 48 inches or less to a highest position where this chin rest would be 69 inches or more above the floor.

3.2.1.3 *Leveling.* — The supporting frame shall be provided with a means for leveling the assembled cassette changer if the proper operation of the unit is dependent upon the level position of the cassette shifting mechanism.

3.2.1.4 *Floor mounting.*—The floor mounting of the unit shall be of such a design that no wall jacks or other wall supports are necessary for continued satisfactory operation of the cassette changer.

3.2.1.5 *Wall mounting.*—The wall mounting of the unit shall comply with all requirements of this specification except paragraph 3.2.1.1.

3.2.2 *Shifting mechanism.* — The shifting mechanism shall provide for the shifting of cassettes in a horizontal plane in the performance of stereoscopic radiography of the chest and other anatomical parts commonly employing a cassette changer.

3.2.2.1 *Motor drive.*—The device for shifting the cassettes shall be motor-driven and provided with automatic provisions for actuating the stereoscopic shift of the X-ray tube. The shifting mechanism shall be controlled and actuated from the control cabinet of the X-ray apparatus used in conjunction with the cassette changer.

3.2.2.2 *The cassette accommodation.* — The cassette shifting mechanism shall accommodate two 14- by 17-inch cassettes with the 17-inch dimension of cassette in either the horizontal or vertical position as may be the choice of the operator. In addition, the cassette shifting mechanism shall be adjustable to accommodate cassettes from the standard 8 by 10 inches to 14 by 17 inches, placed lengthwise or crosswise in the unit.

3.2.2.3 *Cassette shift.* — The cassette shift shall be completed within a two second interval without noticeable vibration with each cassette being placed in the same position in relation to the exposure panel. There shall be no possibility of an incomplete cassette shift as long as no power failure has occurred. If power has been interrupted and the cassette shift was not complete, the shift shall be completed upon the restoring of power.

3.2.2.4 *Insertion and removal of cassettes.*— To provide for the insertion and removal of the cassettes from the cassette shifting mechanism, a switch shall be conveniently located in the housing of the cassette shifting mechanism to cause the shifting of the cassette into and out of the loading position of the shifting mechanism, if shifting of the cassette is necessary in the loading of the cassette changer.

3.2.2.5 The housing of the cassette changer shall be designed to prevent the patient or the patient's clothing from being entangled in the shifting mechanism. The housing shall be provided with an exposure panel and radiation protective position for the exposed and/or unexposed film.

3.2.2.5.1 *Radiation protective panels.*—The radiation protective panels shall be provided with sufficient equivalent lead protection to prevent the fogging of a film in a cassette protected by these panels with the central ray of the X-ray tube normal to the exposure panel. To determine

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the amount of equivalent lead protection, the following exposure factors will apply:

(a) Six feet target-to-film distance
(b) 200 MA tube current
(c) 120 KV tube voltage
(d) 4/10 second exposure

3.2.2.5.2 *Exposure panel.* — The exposure panel shall be such as to provide a distance no greater than 0.5 inch from the front of the cassette in the shifting mechanism to the front of the exposure panel.

3.2.2.5.2.1 *Opacity of exposure panel.*—The opacity of the exposure panel to X-radiation shall be no greater than that of 0.5 mm. aluminum using kilovoltages as low as 60 KVP.

3.2.2.5.2.2 *Mounting of exposure panel.* — The exposure panel shall be positioned in such a manner with respect to a 14- by 17-inch cassette in the cassette shifting mechanism that the distance from the chin rest to the upper border of the film in the cassette is not greater than 2 inches. Further, the panel shall be of such design as to permit lateral head and cervical non-bucky radiography with patient seated with his shoulder under the lower edge of the exposure panel. The distance from the lower edge of the exposure panel to the lower edge of a film in the cassette placed in the shifting mechanism shall be no greater than one inch.

3.2.2.5.3 *Locks and scale.*—The housing of the shifting mechanism shall be provided with easily operated locking device for locking the vertical movement of the mechanism as described in 3.2.1.2. A scale calibrated to correspond with the scale indicating target-to-table top distance on the tube stand shall be provided for properly orienting the cassette shifting mechanism with the X-ray tube for chest radiography.

3.2.2.5.4 *Sanitary paper.* — The housing of the shifting mechanism shall be provided with a hanger supporting a roll of sanitary paper in such a position, and of such size to permit covering the exposure panel or the front of the bucky diaphragm with a sanitary covering of paper from the roll. Provisions shall be made for easy application of this paper and tearing off sections of the paper when used. Each cassette changer will be furnished with six rolls of this paper.

3.2.2.5.5 *Counterbalance.*—The cassette changer shall be perfectly counterbalanced when the cassette shifting mechanism is loaded with two 14- by 17-inch cassettes, half-roll of sanitary paper, and has its locks released, or the vertical movement of cassette shifting mechanism shall be of an easily operated self-retaining hand operated rack and gear type.

3.2.3 *Compression device.*—Attached to the housing covering the shifting mechanism, a compression and immobilizing device shall be incorporated. This shall consist of a strong X-ray transparent fabric belt with suitable fixtures for anchorage. The effective use of this device shall be possible whether or not the bucky diaphragm is used in a radiographic examination. A ratchet type of mechanism shall be part of this device and shall provide for the tightening of the band and for the instant release of its tension. Two of the X-ray transparent fabric belts shall be furnished with each cassette changer.

3.2.4 *Bucky diaphragm.* — The bucky diaphragm supplied shall be as indicated in the bid invitation and shall be of the following focal film distances:

Type I.—48-inch focal film distance.

Type II.—72-inch focal film distance.

3.2.4.1 *Grid movement.* — The mechanism for moving the bucky grid shall be the type commonly known as the reciprocating type in which the movement of the grid is electrically interlocked with the exposure timer of the X-ray control cabinet determining the exposure time. The movement of the grid shall permit grid free radiographs of 1/20 second exposure.

3.2.4.2 *Grid ratio.* — The ratio of the grid shall be 8:1.

3.2.5 *Photo-timer receiver.*—When specified in the bid invitation, the housing of the cassette shifting mechanism shall be provided with a photo-timer receiver. This receiver shall be possible of inter-connection, with photo-timer cir-

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cuits in the control cabinet. The receiver shall be so positioned as to permit correctly exposed photo-timer radiographs of the chest using either non-bucky or bucky radiographic techniques.

3.2.6 *Radiographic grid (types I and II, class 3 only).*—The grid supplied shall be mounted to the cassette changer in such a manner that it can be slid into or out of the radiographic position. When "out-of-the radiographic-position" the grid shall be protected from possible physical or mechanical damage.

3.2.6.1 *Grid characteristics.*—The grid shall be of commercial design having 80 lines per inch, a ratio of 8:1 and shall be free of imperfections. A radiograph of the grid shall reveal no areas of uneven density.

4. TESTS

4.1 The unit shall be tested for compliance with this specification.

5. PREPARATION FOR DELIVERY

5.1 Packing.—The subject commodity shall be prepared for shipment to permit acceptance by carrier for transportation at the lowest applicable rate, and to afford maximum protection from normal hazards of transportation. Unless otherwise specified, shipping containers shall conform to the Uniform Freight Classification Rules in effect on the date of shipment.

5.2 Marking.

5.2.1 *Shipping containers.*—Unless otherwise specified, shipping containers shall be marked with the name, type, class, and style of the commodity, the name of the manufacturer or contractor, the number of the contract or order, the name and address of the consignee, and the gross weight.

6. NOTES

6.1 Ordering data.—Purchasers should specify type, class and style of item required and should exercise any desired options offered herein. (See 1.2.1, 1.2.2, 3.2.4, 5.1, 5.3.1, 6.2 and 7.2.3.)

6.2 It is believed that this specification adequately describes the characteristics necessary to secure the desired material, and that normally no samples will be necessary prior to award to determine compliance with this specification. If, for any particular purpose, samples with bids are necessary, they should be specifically asked for in the invitation for bids, and the purpose to be served by the bid sample should be definitely stated, the specification to apply in all other respects.

6.3 Federal specifications do not include all types, classes, styles, etc., of the commodities indicated by the titles of the specification, or which are commercially available, but are intended to cover the types, etc., which are suitable for Federal Government requirements.

6.4 This specification covers the following item appearing in the Armed Services Medical Stock List:

6525–601–1700 Cassette Changer, 110 Volt, 60 Cycle, a.c.

6.5 Transportation description.—The transportation description applicable to these items is:

Machines and machinery, not otherwise indexed by name.

Carload weight, 24,000 pounds, subject to Rule 34 Uniform Freight Classification.

Truckload weight, 24,000 pounds.

Patent notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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7. DEPARTMENTAL REQUIREMENTS

7.1 The following specifications and standard, of the issues in effect on date of invitation for bids, and special requirements, form a part of this specification for the purchases made under this specification by the respective departments.

7.2 Army, Navy and Air Force.

7.2.1 *Applicable specifications and standard.*

Federal Specifications:

NN–B–621—Boxes, Wood, Nailed and Lock-Corner.

PPP–B–601 — Boxes, Wood, Cleated-Plywood.

Military Specifications:

JAN–P–106—Packaging and Packing for Overseas Shipment—Boxes; Wood, Nailed.

JAN–P–125—Packaging and Packing for Overseas Shipment — Barrier-Materials, Waterproof, Flexible.

Military Standard:

MIL–STD–129—Marking for Shipment and Storage.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions, should be obtained from the procuring activity or as directed by the contracting officer.)

7.2.2 *Packaging.*

7.2.2.1 *Unit of issue.*—One cassette changer, as specified, constitutes one unit.

7.2.2.2 *Unit packaging.*—Each unit shall be disassembled as far as is practicable. All exterior finished surfaces shall be protected by wrapping with kraft paper and cushioning with excelsior pads or flexible corrugated fiberboard or other suitable materials. Wrapping and cushioning material shall be secured by tying with cotton cord or tape, or by sealing with kraft gummed paper tape. Work shall be accomplished in such a manner as to insure against marring, scratching, or damage to the unit.

7.2.3 *Packing.*

7.2.3.1 *For domestic shipment.*—Each unit shall be overpacked in an exterior container constructed in accordance with the requirements of Federal Specification NN–B–621 or PPP–B–601, table I. Plywood shall be type III, class 1. The unit shall be adequately secured within the exterior container by means of bolts, wood braces, block or other suitable devices, to insure against shifting and breakage, or marring of surfaces.

7.2.3.2 *For overseas shipment.*—Each unit shall be overpacked in an exterior container designed for type 3 load, and constructed in accordance with the requirements of Military Specification JAN–P–106 or Federal Specification PPP–B–601, table II. Plywood shall be type III, class 1. Closure and strapping shall comply with appendix of applicable specification. Each box shall be lined with a waterproof barrier shroud fabricated of material conforming to the requirements of Military Specification JAN–P–125. The unit shall be adequately secured within the exterior container by means of bolts, wood braces, blocks or other suitable devices, to insure against shifting and breakage, or marring of surfaces.

7.2.4 *Marking.*

7.2.4.1 *Interior packages.*—Each container shall be marked to identify contents.

7.2.4.2 *Shipping container.* — Shipping container shall be marked in accordance with Military Standard MIL–STD–129.

7.2.5 *Ordering data.*—Requests, requisitions, schedules, and contracts or orders should specify whether the subject commodity is to be packed for domestic or overseas shipment. (See 7.2.3.1 and 7.2.3.2.)

MILITARY INTEREST:

Army—M
Navy—Md
Air Force.

Copies of this specification may be purchased for 10 cents.

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